Multiple-server system

- In multiple-server models, two or more independent servers in parallel server a single waiting line.
- slightly more complex than the single-server queuing system
- Examples
  (i) check in counters at the airport.
  (ii) check out counters in the groceries.

Assumptions of the multiple-server model:

The following assumptions apply in a multi-server mode.

(i) An infinite calling population.
(ii) A first-come, first-served queue discipline.
(iii) poisson arrival rate.
(iv) Exponential service time.

Helpful Formula in queuing analysis:

For multiple-server system:

The parameters of the multiple-server model are as follows:

λ = The arrival rate (average no. of arrivals per time period)

μ = The service rate (average no. of served per time period) per server.

c = The no. of servers = S

$c\mu$ = The mean effective service rate for the system, which must exceed the arrival rate.

(1) The probability that no customers are in the queuing system (either in the queue or being served) is

$$P_0 = \frac{1}{\left[\sum_{n=0}^{c-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n\right] + \left[\frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c\left(\frac{c\mu}{c\mu-\lambda}\right)\right]}$$

Where $c = S$ are same symbols.

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(2) The probability that n customers are in the queuing system is

$$P_n = \frac{1}{n}\left(\frac{\lambda}{\mu}\right)^n P_0 \quad ; if\ n \le c$$

(3) The average no. of customers in the queuing system (i.e., the customers being serviced and in the waiting line) is

$$L_s = \frac{\lambda\mu\left(\frac{\lambda}{\mu}\right)^c}{(c-1)!\,(c\mu-\lambda)^2} P_0 + \frac{\lambda}{\mu}$$

(4) The average no. of customers in the waiting line is

$$L_q = L_s - \frac{\lambda}{\mu}$$

(5) The average time a customer spends in the total queuing system (i.e., waiting and being served) is

$$W_s = \frac{L_s}{\lambda} = W_q + \frac{1}{\mu} = \frac{L_q}{\lambda} + \frac{1}{\mu}$$

(6) The average time a customer spends in the queue to be served is

$$W_q = \frac{L_q}{\lambda}$$

(7) The probability that a customer arriving in the system must wait for service (i.e., the probability that all the servers are busy)

$$P_w = \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \frac{c\mu}{c\mu-\lambda} P_0$$

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For example: 2f

$\lambda = 10$ customers per hour arrived at checkout counter.

$\mu = 4$ customers per hour can be checked out

$c = 3$ servicing counters.

Now,

$$P_0 = \frac{1}{\left[\sum_{n=0}^{c-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)\right] + \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \left(\frac{c\mu}{c\mu-\lambda}\right)}$$

$$= \frac{1}{\left[\frac{1}{0!}\left(\frac{10}{4}\right)^0 + \frac{1}{1!}\left(\frac{10}{4}\right)^1 + \frac{1}{2!}\left(\frac{10}{4}\right)^2\right] + \frac{1}{3!}\left(\frac{10}{4}\right)^3\left(\frac{3\times4}{3\times4-10}\right)}$$

$= 0.045$

$$L_s = \frac{\lambda\mu\left(\frac{\lambda}{\mu}\right)^c}{(c-1)!\,(c\mu-\lambda)^2} P_0 + \frac{\lambda}{\mu}$$

$$= \frac{10\times4\left(\frac{10}{4}\right)^3}{(3-1)!\,(3\times4-10)^2} \times 0.045 + \frac{10}{4}$$

$= 6$ customers in the queuing system.

$L_q = L_s - \frac{\lambda}{\mu} = 6 - \frac{10}{4} = 3.5$ customers waiting to be served.

$W_s = \frac{L_s}{\lambda} = \frac{6}{10} = 0.60$ hours $= 36$ minutes.

$W_q = \frac{L_q}{\lambda} = \frac{3.5}{10} = 0.35$ hours $= 21$ minutes.

<u>Question:</u> A nationalised telephone company has 4 service counters at its service center providing services related to new connection, renewal, bill-pay, resolving billing problem etc. Average arrivals are 12 customers/hr in an 8 hr working day. service time varies from customer to customers, but on an average its 15 minutes/customers. Now calculate:

(i) Average number of customers in the system.

(ii) Average no. of customers waiting for service.

(iii) Average time a customer spends in the system.

(iv) Average waiting time per customers.

(v) probability that customer has to wait to get service.

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Solution: Given,

$\lambda = 12$ customers/hr

$\mu = 4$ customers/hr.

$c = 4$ (no. of channels)

Now,

Probability that no customers in the system is

$$P_0 = \left[\sum_{c=0}^{c-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \cdot \left(\frac{c\mu}{c\mu-\lambda}\right) \right]^{-1}$$

$$= \left[\frac{1}{0!}\left(\frac{12}{4}\right)^0 + \frac{1}{1!}\left(\frac{12}{4}\right)^1 + \frac{1}{2!}\left(\frac{12}{4}\right)^2 + \frac{1}{3!}\left(\frac{12}{4}\right)^3 + \frac{1}{4!}\left(\frac{12}{3}\right)^4 \left(\frac{4\times4}{4\times4-12}\right) \right]^{-1}$$

$$= \left(\frac{53}{2}\right)^{-1} = \frac{2}{53} = 0.0377$$

Then,

(i) Average no. of customers in the system is

$$L_s = \frac{\lambda\mu\left(\frac{\lambda}{\mu}\right)^c}{(c-1)!\,(c\mu-\lambda)^2} P_0 + \frac{\lambda}{\mu}$$

$$= \frac{12\times4\left(\frac{12}{4}\right)^4}{3!\,(4\times4-12)^2} \times 0.0377 + \frac{12}{4}$$

$= 4.52$ customers in the queue system.

(ii) Average no. of customers in waiting line is

$$L_q = L_s - \frac{\lambda}{\mu} = 4.52 - \frac{12}{4} = 1.52 \text{ customers waiting to be served}$$

(iii) Average no. of customers spends in the system is

$$W_s = \frac{L_q}{\lambda} + \frac{1}{\mu} = \frac{1.52}{12} + \frac{1}{4} = \frac{113}{300} \text{ hrs} = \cancel{22.6 \text{ minutes}}$$

$= 22.6$ minute.

(iv) Average time a customer wait for service in the queue is

$$W_q = \frac{L_q}{\lambda} = \frac{1.52}{12} = 0.126 \text{ hr} = 7.6 \text{ min.}$$

(v) probability that customer has to wait:

$$P(n \geq c) = \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \left(\frac{c\mu}{c\mu-\lambda}\right) P_0$$

$$= \frac{1}{4!}\left(\frac{12}{4}\right)^4 \cdot \frac{4\times4}{(4\times4-12)} \times 0.0377 = 0.50 \text{ Ans.}$$

M/M/S : N/FCFS or M/M/c/k model.

Multiple server queuing model with finite queue.

(1) The probability of n customers in the system in the steady-state condition is

$$P_n = \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n P_0 \; ; \; 0 \le n < s.$$

$$\frac{1}{s!\, s^{n-s}}\left(\frac{\lambda}{\mu}\right)^n P_0 \; ; \; s \le n \le N.$$

(2) system shall be iddle is

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n + \sum_{n=s}^{N} \frac{1}{s!\, s^{n-s}}\left(\frac{\lambda}{\mu}\right)^n\right]^{-1}$$

$$= \left[\sum_{n=0}^{s-1} \frac{(s\rho)^n}{n!} + \frac{1}{s!}\left(\frac{\lambda}{\mu}\right)^s (N-s+1)\right]^{-1}$$

(3) The expected no. of customers in the queue system :

$$L_q = \cancel{L_q +} \sum_{n=s}^{N} (n-s) P_n$$

$$= \frac{(s\rho)^s \rho}{s!\,(1-\rho)^2}\left[1-\rho^{N-s+1} - (1-\rho)(N-s+1)\rho^{N-s}\right] P_0$$

(4) The expected no. of customers in the system :

$$L_s = L_q + \frac{\lambda(1-P_N)}{\mu}$$

$$= L_q + s - P_0 \sum_{n=0}^{s-1} \frac{(s-n)}{n!}\left(\frac{\lambda}{\mu}\right)^n.$$

(5) The expected waiting time in the system:

$$W_s = \frac{L_s}{\lambda(1-P_N)}$$

(6) The expected waiting time in the queue :

$$W_q = W_s - \frac{1}{\mu} = \frac{L_q}{\lambda(1-P_N)}$$

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Question: Let there be an automobile inspection situation with three inspection stalls. Assume that cars waits in such a way that when stall; becomes vacant, the car at the head of the line pulls up to it. The station can accommodate almost seven cars at one time. The arrival pattern is poisson with a mean of one car every minute during the peak hour. The service time is exponential with mean of 6 minutes

(i) Find the average no. of customers in queue during the peak hours?

(ii) Find the average number of customers in the system during peak hour?

(iii) Expected waiting time in queue and in the system?

(iv) Average no. of cars per hour that cannot inter the situation because of full capacity.

Solution: Given,

$$S = C = 3, \quad N = 7, \quad \lambda = 1 \text{ car/min}, \quad \mu = \frac{1}{6} \text{ car/min}.$$

Now,

$$P_0 = \frac{1}{1141}$$

Then

(i) $$L_q = \sum_{n=3}^{7} (n-s) P_n = \sum_{n=3}^{7} (n-s) \cdot \frac{1}{s!\, s^{n-s}} \left(\frac{\lambda}{\mu}\right)^n P_0 = 3.09 \text{ cars}.$$

(ii) $$L_s = L_q + \frac{\lambda(1-P_N)}{\mu} = L_q + 6(1-P_N) = 3.09 + 3 = 6.09 \text{ cars}.$$

(iii) $$W_s = \frac{L_s}{\lambda(1-P_N)} = 12.3 \text{ min}.$$

$$W_q = W_s - \frac{1}{\mu} = 12.3 - 6 = 6.3 \text{ min}.$$

(iv) $$1 \text{ hr} \times \lambda \times P_N = 60 \text{ min} \times \lambda \times P_N = 60 \times 1 \times \frac{(6)^7}{3!\, 3^4} \times \frac{1}{1141}$$

$$= 30.4 \text{ cars/hr}.$$

(The expected no. of cars per hour that cannot entr the station is

$1 \text{ hr} \times \lambda \times P_N$)